QAOA Maxcut currently failing on the latest version of PennyLane and aut

https://github.com/PennyLaneAI/qml/issues/36

ROW 110

QAOA Maxcut currently failing on the latest version of PennyLane and autograd #36

New issue

Closed

josh146 opened on Feb 8, 2020

The QAOA maxcut tutorial is currently failing with the following traceback:

```
  File "implementations/tutorial_qaoa_maxcut.py", line 208, in circuit
    U_C(gammas[i])
IndexError: index 1 is out of bounds for axis 0 with size 1
```

The issue seems to be passing the parameter `gamma`. At line 246,

```
neg_obj -= 0.5 * (1 - circuit(gammas, betas, edge=edge, n_layers=n_layers))
```

it can be seen that `gamma = Autograd ArrayBox with value [0.00617866 0.00415137]`. However, within the QNode at line 202,

```
@qml.qnode(dev)
def circuit(gammas, betas, edge=None, n_layers=1):
    # apply Hadamards to get the n qubit |+> state
    [print(2, i.val) for i in gammas if isinstance(i, qml.variable.Variable)]
```

we have `gamma = [0.006178660018330691]` - the second element is 'missing'.

josh146 added bug on Feb 8, 2020

antalszava on Feb 11, 2020 — Contributor

The problem seems to be related to the fact that there is only one `QNode` that is being created throughout the entire execution of the script.

When calling `_construct` with two parameters for gamma, the same `self.arg_vars` will be used as created for only one parameter (as it is `not None`), yielding the error.

A quick way to check this would be commenting out lines `bitstrings1 = qaoa_maxcut(number_of_layers=1)[1]` and `plt.hist(bitstrings1, bins=bins)`.

Going forward with trying to figure out why there's only one `QNode` that is being created.

antalszava on Feb 11, 2020 — Contributor

With one `QNode` the logic should be altered here such that there is a more in-depth check comparing the current `self.arg_vars` and the new `Variables`:
https://github.com/XanaduAI/pennylane/blob/719e27c7d046e7d488b8c68685fe803f4f5afa9f/pennylane/qnodes/base.py#L500

josh146 on Feb 11, 2020 — Member — Author

Thanks @antalszava! Was this bug introduced with the circuit drawer then? We should add a test to PL to test this specific edge case, and make sure it is fixed.

👍 1

Assignees
No one assigned

Labels
bug

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
BaseQNode._construct fix
PennyLaneAI/pennylane

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Participants

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antalszava on Feb 11, 2020 Contributor

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Participants

antalszava on Feb 11, 2020 Contributor

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antalszava mentioned this on Feb 12, 2020

Fix multiple evaluations of mutable QNode PennyLaneAI/pennylane#505

Include unit tests for auxiliary methods of BaseQnode PennyLaneAI/pennylane#506

smite mentioned this on Feb 14, 2020

BaseQNode._construct fix PennyLaneAI/pennylane#510

antalszava on Feb 18, 2020 Contributor

With PennyLaneAI/pennylane#505 merged, I'll close this for now.

antalszava closed this as completed on Feb 18, 2020

Fix multiple evaluations of mutable QNode #505

Code

Merged josh146 merged 18 commits into `master` from `mutable_qnode_make_variable` on Feb 18, 2020

Conversation 31 · Commits 18 · Checks 0 · Files changed 3 · +57 −3

antalszava commented on Feb 12, 2020 • edited · Contributor

Context:
Currently, a mutable `QNode` evaluated multiple times fails if there is a change in the list `Variable` instances that are being used. This can be seen currently when running the QAOA Maxcut tutorial.

Description of the Change:
Adds a slightly more sophisticated check for when to generate the list of `Variable` instance in the `BaseQNode._construct` method.

Benefits:
A very lightweight check is being used for mutable `QNode`. This assumes that the input `args` argument can be converted to a `numpy.ndarray` and that the `shape` attribute of this `numpy.ndarray` will uniquely identify the arguments passed to `_construct`. The use of methods from the `inspect` library is deliberately avoided such that no considerable overhead is introduced.

The test suite passes including a new Integration test getting the edge case similar to the one in the QAOA MaxCut tutorial.

Possible Drawbacks:
This PR addresses the case of when the `QNode` is mutated in its args in a specific variety of ways (querying the shape of a `numpy.ndarray`).

Further TODO items are contained in the following issues:
#506 (this might also contain an unexpected behaviour when using keyword arguments)
#507

Related GitHub Issues:
PennyLaneAI/qml#36

antalszava added 5 commits 5 years ago

First integration test (catches the index out of range error for muta…	d553ea1
Revert logic and Qnode Base integration test	7c77571
Add args_changed check for mutable QNodes	b277b92
Modify construction condition	8862e88
Docstring	1549cb8

Reviewers
josh146
johannesjmeyer
smite

Assignees
No one assigned

Labels
bug · review-ready

Projects
None yet

Milestone
No milestone

Development
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